

The function

$$f(x_1, x_2) = 400 - 3x_1^2x_2^2$$

is differentiable at  $(1, -2)^T$ . Therefore the gradient is the direction of steepest ascent. We have

$$\nabla f(x_1, x_2) = \begin{pmatrix} -6x_1x_2^2 \\ -6x_1^2x_2 \end{pmatrix},$$

and

$$\nabla f(1, -2) = \begin{pmatrix} -24 \\ 12 \end{pmatrix}.$$

As  $\|\nabla f(1, -2)\| = 12\sqrt{5}$ , the normalized direction in which the depth increases as fast as possible is then

$$d = \left( -2/\sqrt{5}, 1/\sqrt{5} \right).$$

If she swims one meter along the direction, her new position is

$$x^+ = \begin{pmatrix} 1 - 2/\sqrt{5} \\ -2 + 1/\sqrt{5} \end{pmatrix},$$

and the depth at this position is

$$f(x^+) = 400 - 3 \frac{269 - 120\sqrt{5}}{25} \approx 399.92.$$

If she swims a distance of 8 meters, her new position is

$$x^{++} = \begin{pmatrix} 1 - 16/\sqrt{5} \\ -2 + 8/\sqrt{5} \end{pmatrix},$$

and the depth is

$$f(x^{++}) \approx 117.$$

As the depth at the current position is

$$f(x) = 388,$$

it illustrates that following an ascent direction increases the value of the function, but up to a point. If the step along the direction is too long, the value of the function may actually decrease. In this example, the step should be lesser than 5.59 in order to obtain an increase of the value of the function.